I Bharath Chandra Nallani declare that I have completed this assignment completely and entirely on my own, without any consultation with others. I understand that any breach of the UAB Academic Honor Code may result in severe penalties.

Signature: Bharath Chandra Nallani

Date: 01/31/2025

Web username: bnallaniweb

Password: 1234

Part A:

1.

```
SELECT e.cno, e.sectno, AVG(s.gpa) AS avg_gpa
FROM enroll e
JOIN student s ON e.sid = s.sid
JOIN course c ON e.cno = c.cno AND e.dname = c.dname
WHERE c.dname = 'Computer Sciences'
GROUP BY e.cno, e.sectno;
```

```
bnallani=> SELECT e.cno, e.sectno, AVG(s.gpa) AS avg_gpa
FROM enroll e
JOIN student s ON e.sid = s.sid
JOIN course c ON e.cno = c.cno AND e.dname = c.dname
WHERE c.dname = 'Computer Sciences'
GROUP BY e.cno, e.sectno;
 cno | sectno |      avg_gpa
-----+--------+--------------------
 302 | 1      |                  3
 302 | 2      |              3.075
 467 | 1      | 2.9799999999999995
 701 | 1      | 3.2833333333333337
 726 | 1      | 2.6411764705882352
(5 rows)
```

2.

```
SELECT c.cname, e.cno, e.sectno
FROM enroll e
JOIN course c ON e.cno = c.cno AND e.dname = c.dname
GROUP BY c.cname, e.cno, e.sectno
HAVING COUNT(e.sid) < 6;
```

```
bnallani=> SELECT c.cname, e.cno, e.sectno
FROM enroll e
JOIN course c ON e.cno = c.cno AND e.dname = c.dname
GROUP BY c.cname, e.cno, e.sectno
HAVING COUNT(e.sid) < 6;
            cname            | cno | sectno
-----------------------------+-----+--------
 Advanced Garbage Collection | 561 | 1
(1 row)
```

3.

```
SELECT DISTINCT m.dname
FROM major m
JOIN student s ON m.sid = s.sid
WHERE s.age < 18;
```

```
bnallani=> SELECT DISTINCT m.dname
FROM major m
JOIN student s ON m.sid = s.sid
WHERE s.age < 18;
         dname
------------------------
 Industrial Engineering
 Mathematics
(2 rows)
```

4.

```
SELECT DISTINCT s.sname, COALESCE(m.dname, '') AS major
FROM enroll e
JOIN student s ON e.sid = s.sid
JOIN course c ON e.cno = c.cno AND e.dname = c.dname
LEFT JOIN major m ON s.sid = m.sid
WHERE c.cname LIKE '%College Geometry%';
```

```
bnallani=> SELECT DISTINCT s.sname, COALESCE(m.dname, '') AS major
FROM enroll e
JOIN student s ON e.sid = s.sid
JOIN course c ON e.cno = c.cno AND e.dname = c.dname
LEFT JOIN major m ON s.sid = m.sid
WHERE c.cname LIKE '%College Geometry%';
       sname        |        major
--------------------+----------------------
 Atny, Mary H.      | Civil Engineering
 Austin, G.         | Chemical Engineering
 Bates, Michael L.  |
 Cheong, R.         | Computer Sciences
 Davis, Scott P.    |
 Dunbar, D.         | Civil Engineering
 Ford, Gerald       | Chemical Engineering
 Ghandi, I.         | Mathematics
 Glitch, R.         | Civil Engineering
 Gooch              | Computer Sciences
 Mathews, John W.   | Chemical Engineering
 Rosemeyer, S.      | Civil Engineering
 Smith, L.          | Computer Sciences
 Sulfate, Barry M.  | Computer Sciences
 Sulfate, Barry M.  | Sanitary Engineering
 Thorton, James Q.  | Computer Sciences
 Uoiea, Z.          | Mathematics
 Zappa, F.          | Mathematics
 Ziebart, F.        | Civil Engineering
(19 rows)
```

5.

```
SELECT d.dname, d.numphds
FROM dept d
WHERE d.dname NOT IN (
   SELECT DISTINCT m.dname
   FROM major m
   JOIN enroll e ON m.sid = e.sid
   JOIN course c ON e.cno = c.cno AND e.dname = c.dname
   WHERE c.cname LIKE '%College Geometry%'
);
```

```
bnallani=> SELECT d.dname, d.numphds
FROM dept d
WHERE d.dname NOT IN (
    SELECT DISTINCT m.dname
    FROM major m
    JOIN enroll e ON m.sid = e.sid
    JOIN course c ON e.cno = c.cno AND e.dname = c.dname
    WHERE c.cname LIKE '%College Geometry%'
);
         dname          | numphds
------------------------+---------
 Industrial Engineering |      41
(1 row)
```

6.

```
SELECT DISTINCT s.sname
FROM student s
JOIN enroll e1 ON s.sid = e1.sid
JOIN course c1 ON e1.cno = c1.cno AND e1.dname = c1.dname
JOIN enroll e2 ON s.sid = e2.sid
JOIN course c2 ON e2.cno = c2.cno AND e2.dname = c2.dname
WHERE c1.dname = 'Computer Sciences'
AND c2.dname = 'Mathematics';
```

```
bnallani=> SELECT DISTINCT s.sname
FROM student s
JOIN enroll e1 ON s.sid = e1.sid
JOIN course c1 ON e1.cno = c1.cno AND e1.dname = c1.dname
JOIN enroll e2 ON s.sid = e2.sid
JOIN course c2 ON e2.cno = c2.cno AND e2.dname = c2.dname
WHERE c1.dname = 'Computer Sciences'
AND c2.dname = 'Mathematics';
   sname
-----------
 Zappa, F.
(1 row)
```

7.

```
SELECT MAX(s.age) - MIN(s.age) AS age_difference
FROM student s
JOIN major m ON s.sid = m.sid
WHERE m.dname = 'Computer Sciences';
```

```
bnallani=> SELECT MAX(s.age) - MIN(s.age) AS age_difference
FROM student s
JOIN major m ON s.sid = m.sid
WHERE m.dname = 'Computer Sciences';
 age_difference
----------------
             38
(1 row)
```

8.

```
SELECT m.dname, AVG(s.gpa) AS avg_gpa
FROM major m
JOIN student s ON m.sid = s.sid
WHERE s.gpa < 1.0
GROUP BY m.dname;
```

```
bnallani=> SELECT m.dname, AVG(s.gpa) AS avg_gpa
FROM major m
JOIN student s ON m.sid = s.sid
WHERE s.gpa < 1.0
GROUP BY m.dname;
         dname          | avg_gpa
------------------------+---------
 Civil Engineering      |       0
 Computer Sciences      |     0.7
 Industrial Engineering |    0.35
(3 rows)
```

9.

```
SELECT s.sid, s.sname, s.gpa
FROM student s
WHERE NOT EXISTS (
   SELECT c.cno
   FROM course c
   WHERE c.dname = 'Civil Engineering'
   EXCEPT
   SELECT e.cno
   FROM enroll e
   WHERE e.sid = s.sid AND e.dname = 'Civil Engineering'
);
```

```
bnallani=> SELECT s.sid, s.sname, s.gpa
FROM student s
WHERE NOT EXISTS (
    SELECT c.cno
    FROM course c
    WHERE c.dname = 'Civil Engineering'
    EXCEPT
    SELECT e.cno
    FROM enroll e
    WHERE e.sid = s.sid AND e.dname = 'Civil Engineering'
);
 sid |     sname      | gpa
-----+----------------+-----
 29  | Hamilton, S. | 2.8
(1 row)
```

PART B:

2.

```
bnallani=> CREATE TABLE Product (maker VARCHAR(100), model VARCHAR(50) PRIMARY KEY, type VARCHAR(50));
CREATE TABLE
bnallani=> CREATE TABLE PC (model VARCHAR(50) PRIMARY KEY, speed FLOAT, ram INT, hd INT, price FLOAT, FOREIGN KEY (model) REFERENCES Product(model));
CREATE TABLE
bnallani=> CREATE TABLE Laptop (model VARCHAR(50) PRIMARY KEY, speed FLOAT, ram INT, hd INT, screen FLOAT, price FLOAT, FOREIGN KEY (model) REFERENCES Product(model));
CREATE TABLE
bnallani=> CREATE TABLE Printer (model VARCHAR(50) PRIMARY KEY, color VARCHAR(10), type VARCHAR(50), price FLOAT, FOREIGN KEY (model) REFERENCES Product(model));
CREATE TABLE
bnallani=> INSERT INTO Product (maker, model, type) VALUES ('D', 'D001', 'PC'), ('D', 'D002', 'Laptop'), ('A', 'A101', 'PC'), ('A', 'A102', 'Printer'), ('B', 'B201', 'PC'), ('B', 'B202', 'P
rinter'), ('C', 'C301', 'Laptop');
INSERT 0 7
bnallani=> INSERT INTO PC (model, speed, ram, hd, price) VALUES ('D001', 3.5, 16, 512, 1200), ('A101', 3.0, 8, 256, 800), ('B201', 2.8, 16, 1024, 1500);
INSERT 0 3
bnallani=> INSERT INTO Laptop (model, speed, ram, hd, screen, price) VALUES ('D002', 3.2, 16, 512, 15.6, 1400), ('C301', 2.5, 8, 256, 14.0, 1000);
INSERT 0 2
bnallani=> INSERT INTO Printer (model, color, type, price) VALUES ('A102', 'Yes', 'Laser', 300), ('B202', 'No', 'Inkjet', 150);
INSERT 0 2
```

A. SELECT AVG(price) AS avg_price FROM (SELECT price FROM PC WHERE model IN (SELECT model FROM Product WHERE maker = 'D') UNION ALL SELECT price FROM Laptop WHERE model IN (SELECT model FROM Product WHERE maker = 'D')) AS combined_prices;

```
bnallani=> SELECT AVG(price) AS avg_price FROM (SELECT price FROM PC WHERE model IN (SELECT model FROM Product WHERE maker = 'D') UNION ALL SELECT price FROM Laptop WHERE model IN (SELECT m
odel FROM Product WHERE maker = 'D')) AS combined_prices;
 avg_price
-----------
      1300
(1 row)
```

b.

SELECT AVG(pc.hd) AS avg_hd_size FROM PC pc JOIN Product p1 ON pc.model = p1.model WHERE p1.maker IN (SELECT DISTINCT p2.maker FROM Product p2 JOIN Printer pr ON p2.model = pr.model);

```
bnallani=> SELECT AVG(pc.hd) AS avg_hd_size FROM PC pc JOIN Product p1 ON pc.model = p1.model WHERE p1.maker IN (SELECT DISTINCT p2.maker FROM Product p2 JOIN Printer pr ON p2.model = pr.mo
del);
     avg_hd_size
---------------------
 640.0000000000000000
(1 row)

bnallani=>
```